



Highly efficient Ru/B₄C multifunctional oxygen electrode for rechargeable Li–O₂ batteries

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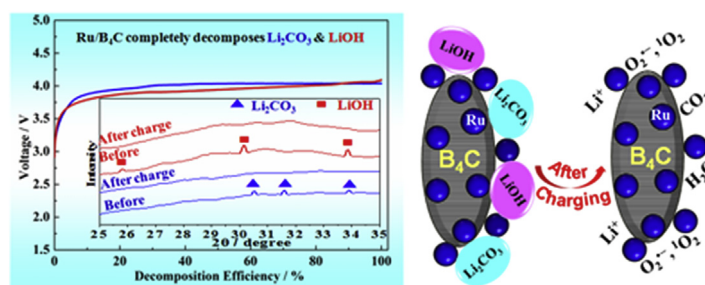
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HIGHLIGHTS

- Non-carbon-based Ru/B₄C oxygen electrode shows good activity for both ORR and OER.
- Ru/B₄C can completely decompose Li₂CO₃ and LiOH below 4.1 V.
- Ru/B₄C is very stable against superoxide radical anion and Li₂O₂.
- Li–O₂ batteries using Ru/B₄C oxygen electrodes exhibit long cycling stability.
- Side reaction during charge accounts for degradation of Ru/B₄C oxygen electrode.

GRAPHICAL ABSTRACT



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ABSTRACT

Irreversible parasitic reactions and the resulting accumulation of insulating side products are main barriers for practical application of rechargeable lithium–oxygen (Li–O₂) batteries. Therefore, it is critical to develop multifunctional oxygen electrodes suitable for oxygen reduction reaction, oxygen evolution reaction, and decomposition of side reaction products. Here we report the application of ultrafine ruthenium on boron carbide (Ru/B₄C) with highly efficient multifunctional activities as carbon-free oxygen electrodes for Li–O₂ batteries. Li₂CO₃ and LiOH can be completely decomposed by Ru/B₄C at 4.0 V and 4.1 V, respectively, within the stability window of the electrolyte. A Li–O₂ battery using the Ru/B₄C oxygen electrode achieves low overpotentials for Li–O₂ reactions, and excellent cycling performance under the capacities of 300 and 500 mAh g_{Ru/B₄C}^{−1}. In-situ gas chromatography analysis reveals that O₂ is the major gas product during charging. Only a negligible amount of CO₂ is observed in the first charging process. Therefore, Ru/B₄C can be a very promising oxygen-electrode material for Li–O₂ batteries and Li–air batteries operated in ambient air.

1. Introduction

Irreversible parasitic reactions, which arise from the nucleophilic attack of active oxygen species toward the carbon-based oxygen

electrode, Li metal anode, and electrolyte have severely limited the development of Li–O₂ batteries [1,2]. The electrochemical decomposition of the side products, such as Li₂CO₃ and LiOH, requires much higher charge voltage than those for the discharge product (Li₂O₂ or

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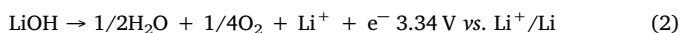
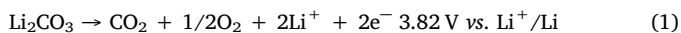
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Li₂O₂), leading to low energy efficiencies [3]. Conventional catalysts are not effective in removing these side products during the recharge process. Redox mediators for oxygen reduction reaction (ORR) [4,5], oxygen evolution reaction (OER) or both [6–9] are effective to reduce the charge/discharge overpotentials in Li–O₂ batteries, but cannot oxidize side products due to their redox potentials lower than those for Li₂CO₃ and LiOH decomposition [10–14]. Therefore, it is highly desirable to have an oxygen electrode with additional catalytic activity for the decomposition of side products. These oxygen electrodes will be particularly beneficial for mediator-aided Li–O₂ batteries or Li–air batteries operated in ambient air since CO₂ and H₂O present in air can lead to more Li₂CO₃ and LiOH formation.

Carbon materials are extensively used for the oxygen electrodes of Li–O₂ batteries, but they are not stable during the discharge and charge process. Non-carbon-based oxygen-electrode materials, such as metal oxides [15], nitrides [16], sulfides [17], noble metals [18,19], and carbides [20–22] possess high inherent stabilities against chemical attack from reactive oxygen species and electrochemical oxidation. Boron carbide (B₄C) has a high resistance against chemical attack and a favorably low density. In our previous work, B₄C powders showed a high inertness when contacted with Li₂O₂ and superoxide (KO₂) powders [21]. In O₂[−] solution, B₄C could be slightly oxidized but the surface oxidation reaction rapidly reached to a steady-state without further deterioration, probably due to the protection of B-suboxides surface layer [21]. As a result, Li–O₂ batteries using B₄C oxygen electrodes achieved more than double the cycles of carbon nanotube (CNT)-based Li–O₂ batteries [21]. Nevertheless, B₄C exhibited poor catalytic capability towards decomposing side products [23]. The typical side products, Li₂CO₃ and LiOH, require high charge voltages to be oxidized, as shown in Eqns. (1) and (2) [3,23].



NiO catalyst was first reported to aid Li₂CO₃ decomposition and subsequently applied in Li–O₂ batteries [24–26]. However, the decomposition efficiency of Li₂CO₃ cannot achieve 100% on NiO [24]. Recently, an ultrafine iridium-decorated boron carbide (Ir/B₄C) nanocomposite has shown excellent multifunction to completely decompose Li₂CO₃ as well as to catalyze ORR and OER for Li–O₂ batteries [23]. But the charge voltage has to be as high as 4.37 V beyond the stability window of organic electrolyte solutions. Accordingly, it is necessary to further increase the activity of the multifunctional catalyst and reduce the decomposition voltage of side products. Ru is a superior OER catalyst and shows even higher activity than Ir [27,28]. In recent years, Ru-based catalysts have been studied in both carbon-based and non-carbon-based oxygen electrodes of nonaqueous Li–O₂ batteries [29–32]. However, their activity on Li₂CO₃ and LiOH decomposition has not been investigated yet. Here we report for the first time the application of Ru/B₄C as a non-carbon-based oxygen-electrode material for aprotic Li–O₂ batteries. The decomposition voltages of Li₂CO₃ and LiOH can be significantly lowered to below 4.1 V, within the stability window of electrolyte, which enables Li–O₂ batteries to achieve excellent cycling performance.

2. Experimental session

2.1. Materials and preparation

B₄C (~50 nm particle size, > 99% purity) and Li₂CO₃ were commercial products purchased from Macklin Biochemical (Shanghai). LiOH was ordered from Aladdin Industrial Co. RuCl₃·xH₂O was ordered from Shanghai Civic Chemical Technology Co. Ltd. Ru/B₄C was synthesized by a modified hydrothermal synthesis method [33]. The Ru-loading in the as-prepared Ru/B₄C catalyst is about 27 wt%, as measured by XPS spectra (Fig. S1). Li₂CO₃- and LiOH-preloaded cells with

B₄C and Ru/B₄C are denoted by B₄C–Li₂CO₃, B₄C–LiOH, Ru/B₄C–Li₂CO₃ and Ru/B₄C–LiOH cells, respectively, and were prepared by a method similar to that in Ref. [23]. Typical loading of Li₂CO₃- or LiOH-preloaded electrodes was 4 mg cm^{−2}. The weight ratio of catalyst/support (B₄C or Ru/B₄C) to reactant (Li₂CO₃ or LiOH) was 4:1. Ru/B₄C oxygen electrodes for Li–O₂ cells were prepared according to the method in Refs. [34,35] with the loading of 0.8–1.2 mg_(Ru/B₄C) cm^{−2}. Li–O₂ cells using Ru/B₄C oxygen electrodes are denoted by Ru/B₄C cells. PVDF was used as the binder and its content was 10% for all types of electrodes. The geometric area of each electrode was 0.5 cm².

2.2. Electrochemical evaluation

Electrochemical cells were assembled as coin cells (CR2032) with holes on positive cases in an argon (Ar) filled glovebox using a Li metal chip as the anode, B₄C- or Ru/B₄C-based electrodes with or without Li₂CO₃ (or LiOH) as the cathode, and 1 M LiTFSI in tetraglyme as the electrolyte. The volume of the electrolyte applied for Li–O₂ cells was 100 μL. The assembled cells were loaded into gas-tight glass containers (with a volume of about 250 cm³). Then the glass containers were transferred out of the glovebox and filled with ultrahigh purity O₂ (1 atm). The Li₂CO₃- or LiOH-preloaded cells were charged in Ar at 0.05 mA cm^{−2} up to the theoretical capacity for decomposing all the Li₂CO₃ or LiOH on a Lanhe battery tester, using 4.2 V as the cutoff voltage, which is within the stability window of tetraglyme-based electrolytes (~4.35 V) [36]. The Ru/B₄C cell was tested in pure O₂ (1 atm) on the Lanhe battery tester at a current density of 0.05 mA cm^{−2} (41.7–62.5 mA g^{−1}) with a cutoff voltage range of 2.0–4.2 V at room temperature, either under the limited capacities of 300 and 500 mAh g^{−1} or in full discharge and charge mode. The cells were equilibrated at open circuit for 4 h before testing. The specific capacity and current are based on the mass of cathode material, which is B₄C or Ru/B₄C. Electrochemical impedance spectroscopy (EIS) of the Ru/B₄C cell was conducted by a CHI 660 electrochemical work station. All impedance data were collected in the frequency range of 1 MHz to 0.1 Hz at open circuit voltage (OCV).

2.3. GC test

The gas released during charging of Li–O₂ cells was analyzed by in-situ GC. In this test, Li–O₂ cells were first assembled in the same setup as the other Li–O₂ cells for discharge/charge tests. The cells were discharged either to a capacity of 300 mAh g^{−1} or to 2.0 V in a glass container at the current density of 0.05 mA cm^{−2}. Then the cells were transferred back to the Ar glovebox and disassembled without any other treatment. The cell components were reassembled using a modified Swagelok cell. The gas outlet of the Swagelok cell was connected to the GC instrument (BeifenGC-8600) by a short polytetrafluoroethylene (PTFE) pipe and the gas inlet was connected to highly pure helium (He) gas, which was used as the sweep gas to purge the charge gas, at a flow rate of 4 ml min^{−1} controlled by mass flow controllers. The charging process was conducted in a Lanhe battery tester alongside the in-situ GC. The emitted charge gas was fed into the GC to analyze the O₂ and CO₂ concentration, and the specific flow rate of an O₂ and CO₂ molecule are regarded to be approximately equal to that of the He sweep gas because the applied current and the quantity of gas product were quite small. Then the specific flow rates of O₂ and CO₂ could be calculated by Eqn. (3).

$$f_{\text{O}_2(\text{CO}_2)} \approx C_{\text{O}_2(\text{CO}_2)} \times f_{\text{He}} \quad (3)$$

Where $f_{\text{O}_2(\text{CO}_2)}$ (μmol min^{−1}) is the O₂ or CO₂ specific flow rate, $C_{\text{O}_2(\text{CO}_2)}$ is the measured O₂ or CO₂ content in the charge gas, and f_{He} is the fixed flow rate of helium gas (4 ml min^{−1}).

2.4. Characterizations

Electrode samples for XRD analysis were tested by a Rigaku D/max-2500 X-ray diffractometer using Cu K α radiation. TEM investigation was conducted using a Hitachi H7650. The morphologies of the electrodes were examined by field-emission SEM (Hitachi S4800). FT-IR and XPS measurements of the electrodes were performed with a Nicolet 6700 and EDAX GENESIS 60S, respectively.

3. Results and discussion

The microstructures of the B₄C and Ru/B₄C nanoparticles (NPs) are shown in Figs. S2 and S3, respectively. The B₄C particles are well crystallized as revealed by HRTEM images and have an average size of ~50 nm (Fig. S2). Ultrafine Ru NPs are well-dispersed (Figs. S3a and b) and tightly deposited (Figs. S3c and d) on the B₄C particle surface, which would facilitate the interactions between Ru catalyst and B₄C substrate and contribute to the synergistic effect [21,27,37]. The specific surface area (SSA) of B₄C particles is about 16.6 m² g⁻¹, as measured by the Brunauer-Emmett-Teller (BET) method. After depositing Ru NPs on B₄C particles, the BET SSA of Ru/B₄C is about 91.2 m² g⁻¹, indicating the significant increase of active sites.

Fig. 1 shows the size of the ultrafine Ru NPs has a narrow distribution (0.75–2.55 nm) and an average size of 1.4 nm (inset of Fig. 1a). The high resolution TEM image in Fig. 1b shows that the ultrafine Ru NPs have well-matched distances of Ru(002) and Ru(100) crystal planes and B₄C particles show a B₄C(003) crystal plane.

3.1. Li₂CO₃ and LiOH decomposition

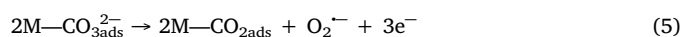
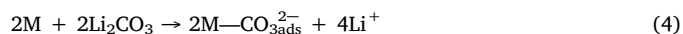
The charge voltage profiles of Ru/B₄C–Li₂CO₃ and Ru/B₄C–LiOH cells are shown in Fig. 2. It shows that the Ru/B₄C–Li₂CO₃ and Ru/B₄C–LiOH cells can fully deliver charge capacities corresponding to the complete decomposition of preloaded Li₂CO₃ and LiOH reactants under very low charge voltages, only 4.0 V and 4.1 V, respectively. Ru/B₄C demonstrates excellent electrocatalytic activities towards Li₂CO₃ and LiOH decompositions within the stability window of electrolyte, which would be conducive to protecting electrolytes from direct electrochemical oxidation and refreshing the electrode surface after each cycle. In contrast, without decoration with ultrafine Ru NPs, B₄C–Li₂CO₃ and B₄C–LiOH cells show rather low charge capacities in Fig. S4a at the cutoff voltage of 4.2 V, corresponding to only about 5.9% and 5.5% of the theoretical capacities for Li₂CO₃ and LiOH decomposition, respectively. Cells using blank B₄C and Ru/B₄C electrodes without reactants of Li₂CO₃ or LiOH were used as blank cells and charged under the same conditions in argon (Ar) to identify the capacity contributed by the potential decomposition of electrolyte. The inset of Fig. 2a and Fig. S4a shows negligible capacities of 0.02 mAh cm⁻² and 0.008 mAh cm⁻² for the B₄C and Ru/B₄C blank cells, respectively, much lower than the capacities for complete decomposition of Li₂CO₃ (0.52 mAh cm⁻²) or LiOH (0.8 mAh cm⁻²). The extremely low charge

capacities of the blank cells reveal that the 1 M LiTFSI-tetraglyme electrolyte is relatively stable below 4.2 V on B₄C and Ru/B₄C.

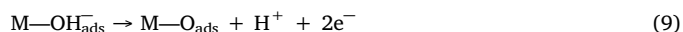
The Li₂CO₃– and LiOH-preloaded Ru/B₄C and B₄C electrodes before and after charging were characterized by XRD to examine the effect of decomposition, as shown in Figs. S4c and d, respectively. The peaks at 44.5, 51.8, and 76.4° correspond to the diffraction from the Ni substrate of the electrode. Fig. 2b shows that Li₂CO₃ and LiOH peaks disappear in the XRD pattern of the after-charging Ru/B₄C sample. Combining this with the results of the charge profiles in Fig. 2a, it is concluded that the Li₂CO₃ and LiOH can be completely decomposed on Ru/B₄C electrode (at least below the detection limit of XRD). To our best knowledge, this is the first report to show successful LiOH decomposition under a low charge voltage by a solid-state catalyst. The XRD patterns in Fig. S4b show that there is no obvious change in the intensity or position of Li₂CO₃ and LiOH diffraction peaks for the B₄C electrode after charging, indicating that Li₂CO₃ and LiOH can hardly be decomposed by pristine B₄C under a relatively low charge voltage.

The possible reaction mechanisms for Li₂CO₃ and LiOH decompositions on Ru/B₄C are suggested in Eqn. (4)–(10), according to the related literature reports [23,37,38].

For Li₂CO₃ decomposition on Ru/B₄C:



For LiOH decomposition on Ru/B₄C:



M represents the active reaction sites on Ru/B₄C catalyst. CO_{3ads} and CO_{2ads} represent the adsorbed carbonate anions and CO₂, respectively. OH_{ads} and O_{ads} represent the adsorbed hydroxyl species and adsorbed oxygen atoms, respectively.

3.2. Li–O₂ cell performance

The Ru/B₄C cell was tested in the full discharge/charge mode between 2.0 and 4.2 V at 0.05 mA cm⁻² for 5 cycles. As shown in Fig. S5a, the Ru/B₄C cell delivers a discharge capacity of 2951 mAh g⁻¹, corresponding to 3.2 mAh cm⁻² in the first cycle. In our previous work, the pristine B₄C cell only delivers a capacity of 900 mAh g⁻¹ in the first full discharge [21]. The increase in discharge capacity for Ru/B₄C cell may be attributed to the increase in reaction active sites by decorating Ru NPs on B₄C. After further adjusting Ru-loading in Ru/B₄C from 27 wt% to 42 wt%, the initial discharge capacity of Ru/B₄C cell can be enhanced to 3993 mAh g⁻¹ (4.3 mAh cm⁻²) (Fig. S5b). A Ru-foam

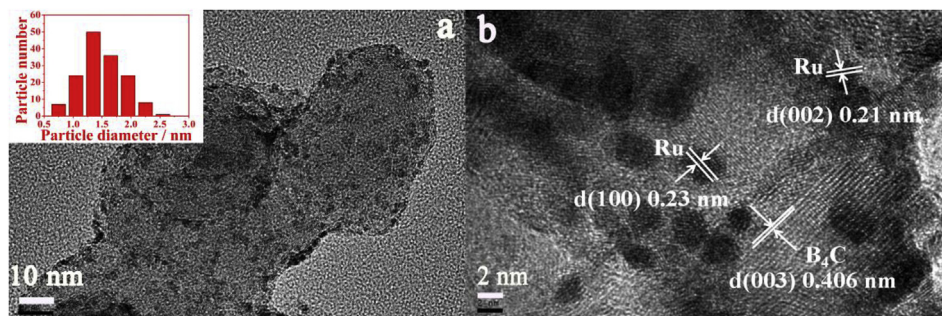


Fig. 1. TEM images of Ru/B₄C NPs under low (a) and high (b) magnifications, which show ultrafine Ru NPs are well dispersed on B₄C NPs.

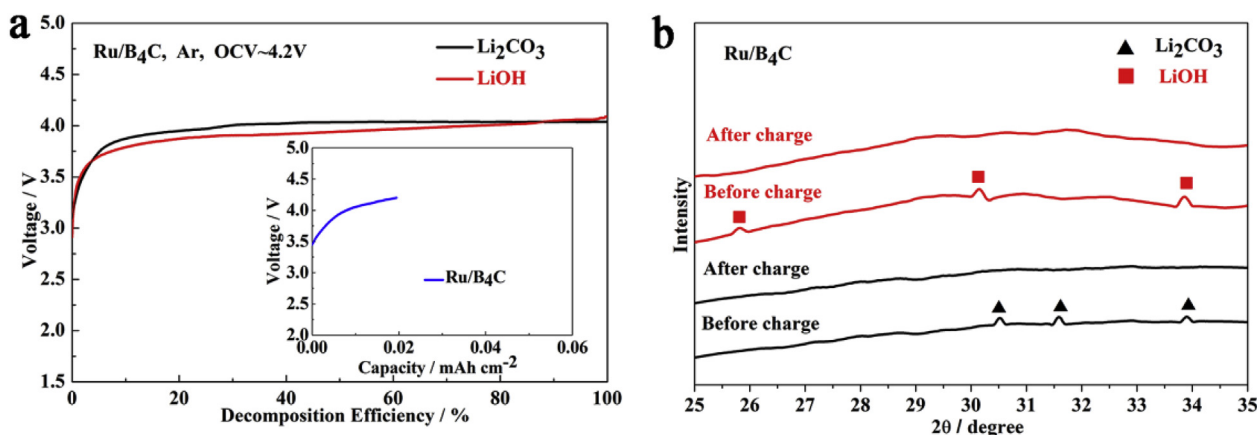


Fig. 2. Charge voltage profiles of Ru/B₄C–Li₂CO₃ and Ru/B₄C–LiOH cells in Ar (a). The inset in (a) shows the charge voltage profiles of the blank Ru/B₄C cell without preloaded Li₂CO₃ and LiOH reactants. Expanded view of the Li₂CO₃ and LiOH X-ray diffraction (XRD) peaks of the Li₂CO₃– and LiOH-preloaded Ru/B₄C electrodes before and after charging (b).

electrocatalyst with an SSA as high as 203.20 m² g^{−1} delivered a full discharge capacity of 2000 mAh g^{−1} at a similar current density to this work (50 mA g^{−1}) [39]. Ru deposited on indium tin oxide (ITO) and Sb-doped tin oxide (STO) also exhibited excellent performances as non-carbon-based oxygen electrodes for Li–O₂ batteries [30,36].

Fig. 3a shows the discharge/charge profiles of the Ru/B₄C cell cycled under a limited capacity of 300 mAh g^{−1} (0.24–0.36 mAh cm^{−2}) at 0.05 mA cm^{−2} within the voltage range of 2.0–4.2 V. The Ru/B₄C cell can deliver the limited discharge and charge capacity for 40 cycles and shows a evident decay after 30 cycles (Fig. 3b). The high-Ru-loading Ru/B₄C (denoted by Ru_(H)/B₄C) cell achieves a more stable cycling under a higher limited capacity of 500 mAh g^{−1}, as shown in Fig. 3c and d. Tested under the same condition at 300 mAh g^{−1}, the Ru_(H)/B₄C cell delivers 100 stable cycles until the battery is voluntarily terminated (Fig. 3e and f).

For all the Ru/B₄C cells, a high discharge voltage plateau of ~2.8 V is achieved in the first cycle, demonstrating the excellent ORR activity of the Ru/B₄C electrode. Impressively, the charge voltage is significantly low, with the terminal voltage at only ~3.7 V in the first charge, which indicates that Ru/B₄C also possesses a very high OER activity. Additionally, the charge voltage profile of the Ru/B₄C cell shows only one charge voltage plateau. In our previous work, two charge voltage plateaus were observed in the charge voltage profile of B₄C cell, where the lower voltage plateau corresponded to the Li₂O₂ oxidation and the higher one was related to the decomposition of side products [21]. One overall charge voltage plateau may indicate the superior catalytic activity of Ru/B₄C for side product decomposition, which reduces the corresponding charge voltage to the range of charge voltage for Li₂O₂ oxidation. The discharge and charge overpotentials are fairly low and show an ascending trend with cycling, suggesting the parasitic reactions still exist and the side products may not be completely removed after each cycle. Corrosion of the oxygen electrode is tentatively ruled out as the B₄C-based oxygen electrode is adequately stable in Li–O₂ batteries [21,23]. Thus, it is most likely that the side reactions arise from oxidation of electrolyte. Considering that the charge voltage of the Ru/B₄C cell has been effectively reduced to below 4.2 V within the stability window of the tetraglyme-based electrolyte, as well as the very small amount of electrolyte oxidation in the charge test of the blank Ru/B₄C cell (the inset in Fig. 2a), the direct electrochemical self-decomposition of the tetraglyme-based electrolyte should be excluded. Chemical or electrochemical attack by the reactive oxygen species towards the electrolyte is probably the root causes of the electrolyte decomposition. In our previous work, the oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) in aprotic Li–O₂ batteries were systematically investigated by using electron paramagnetic resonance (EPR) spectroscopy [40]. The study verified the

formation of the superoxide radical anion (O₂^{•−}) in the ORR during the discharge process, while no O₂^{•−} was detected in the OER during the charge process. Recently, we have further investigated the reaction mechanisms of Li–O₂ batteries using environmental transmission electron microscopy, which revealed that the disproportionation reaction more likely dominates ORR to form hollow spherical Li₂O₂ particles and LiO₂ could also be formed as an intermediate phase during the decomposition of Li₂O₂ in charge process [41]. Some studies using differential electrochemical mass spectroscopy (DEMS) revealed that the e[−]/O₂ ratio deviated significantly from two on charge and more of the side products formed during charging [42,43]. In addition, the parasitic reactions could occur at charging potentials well within the stability window of carbon and electrolyte [13,44]. Since the loading of Li₂CO₃ (0.52 mAh cm^{−2}) or LiOH (0.8 mAh cm^{−2}) in the pre-loaded Ru/B₄C–Li₂CO₃ and Ru/B₄C–LiOH cells far exceeds the limited capacity of 300 mAh g^{−1} (0.24–0.36 mAh cm^{−2}) for the test of Li–O₂ cells, Ru/B₄C should have decomposed all the Li₂CO₃ and LiOH formed during discharge process. The formation and accumulation of side products on Ru/B₄C electrode is likely resulted from the side reactions during charging. A recent work reported that singlet oxygen (¹O₂) accounted for the parasitic reactions and might be formed at charging potentials within the stability window of carbon and electrolyte [10]. We speculate that during the decomposition of Li₂O₂ and the side product on charge some active oxygen species may be generated and trigger the decomposition of electrolyte, which continuously produce Li₂CO₃ and LiOH alongside their decomposition. The terminal discharge and charge voltages are relatively stable and consistent for all the Ru/B₄C cells in the first several tens of cycles, indicating that the Ru/B₄C electrode can effectively refresh the electrode surface during this period. After that, the accumulation of side products may become severe and cause performance decay. Two strategies have been recently applied to address the intractable parasitic reaction issues. One is to develop truly stable electrolytes, such as the inorganic electrolytes (solid [45] or molten salt [46]). Another is to converse the discharge product from highly reactive Li₂O₂ or LiO₂ to benign Li₂O [47].

Fig. S6 shows the impedance spectra of the Ru/B₄C cell in different states. The left-hand (high frequency) intercept of the impedance data with the Z_{re} axis, designated R_b, gives the total ohmic resistance of the cell. The distance between the intercepts at high (left) and low frequency (right), designated R_p, gives the electrode polarization resistance corresponding to the electrocatalytic activity of electrodes. The cell resistance of the Ru/B₄C cell is only 5 Ω cm², which is relatively low among Li–O₂ batteries using non-carbon-based oxygen electrodes. R_b increases a little to 6 Ω cm² after the first discharge due to the formation of insulating Li₂O₂ products. After the first charge, the cell resistance decreases back to 5.3 Ω cm², indicating an excellent capability of Ru/

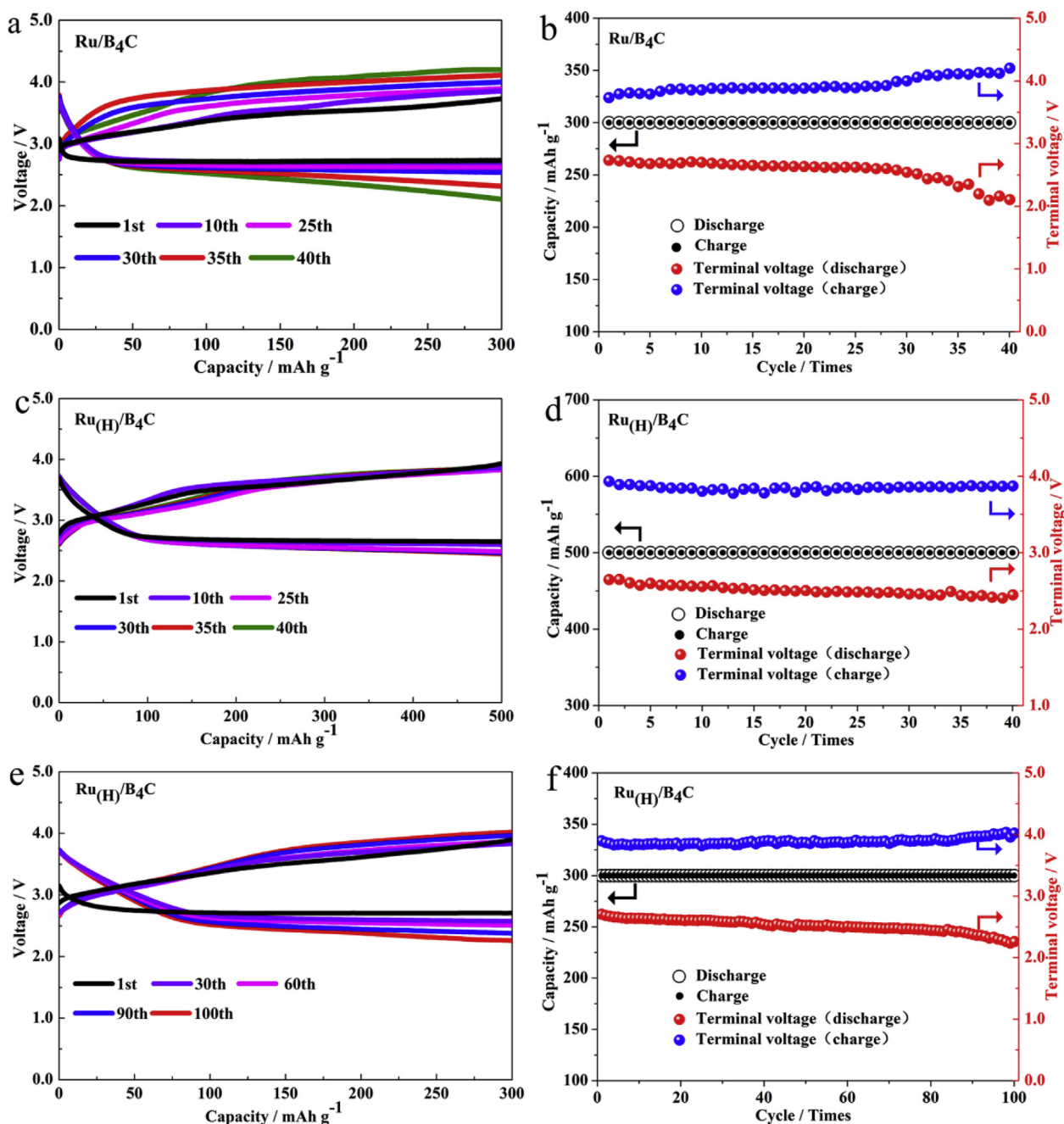


Fig. 3. Discharge/charge voltage profiles of Li–O₂ batteries using Ru/B₄C oxygen electrodes with Ru-loading of 27% (a, b) and 42 wt% (c ~ f) under a limited capacity cycling protocol of 300 mAh g⁻¹ (a, b, e, f) and 500 mAh g⁻¹ (c, d) within the voltage range of 2.0–4.2 V. Ru(H)/B₄C represents the Ru/B₄C oxygen electrode with high-Ru-loading of 42 wt%.

B₄C to recover the electrode surface. After 40 cycles, the R_p increases to 60 Ω cm², implying the accumulation of insulating side products. R_p gradually increases with time, perhaps due to the gradual loss of active sites on the Ru/B₄C electrode. After 40 cycles, R_p increases greatly and the impedance spectra show the typical character of Warburg impedance at low frequency, which represents the polarization of mass transfer. The buildup of insulating side products may cause the blockage of pores in the oxygen electrode and hinder the transfer of O₂ and Li⁺ as well as the electrons.

For the comparison of the Ru/B₄C oxygen electrode with carbon-based oxygen electrode, Ru/Carbon composite was prepared using the same method for the preparation of Ru/B₄C. Graphite (Shanghai Civi-Chem Company) with a similar BET SSA of 17 m² g⁻¹ to B₄C (16.6 m² g⁻¹) was selected as the carbon substrate. Fig. S7a shows the

full discharge/charge voltage profiles for the Li–O₂ cells using Ru/Carbon oxygen electrode between 2.0 and 4.2 V at 0.05 mA cm⁻² for 5 cycles. Comparing Fig. S7a with S5, the initial discharge capacity for Ru/Carbon oxygen electrode is about 2499 mAh g⁻¹, evidently lower than that for the Ru/B₄C electrode. This difference is probably due to the excellent catalytic activity of B₄C for ORR, which may work with Ru catalyst via a synergistic effect [21,23,37]. After the first cycle, the performance of Ru/Carbon electrode decays very fast, attributed to the poor stability of carbon substrate. Figs. S7b and c further show that the Ru/Carbon cell performs inferiorly under limited capacity cycling protocol of 300 mAh g⁻¹.

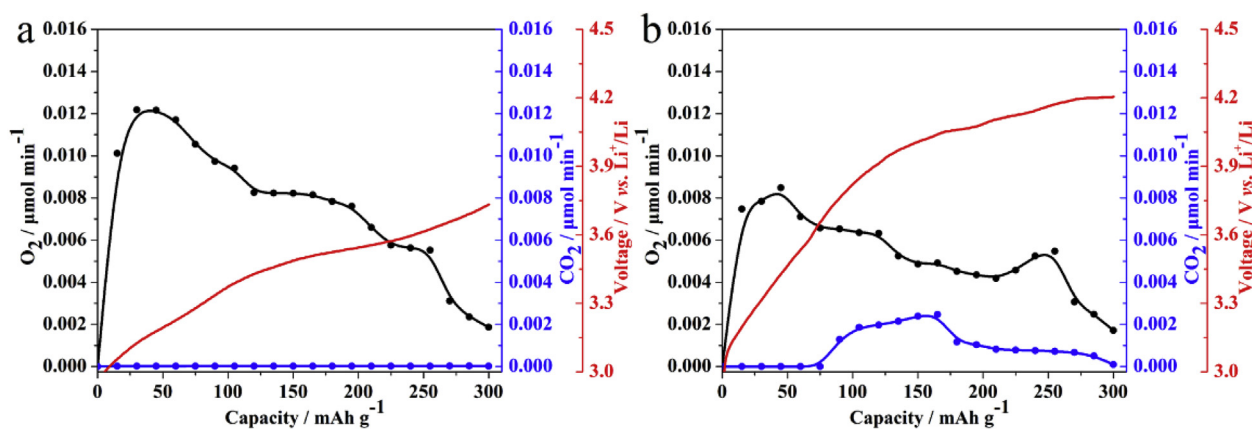


Fig. 4. O₂ and CO₂ gas releases during the first (a) and the 40th (b) charge for the Ru/B₄C cell under the limited capacity of 300 mAh g⁻¹.

3.3. Charge gas products analysis

Real-time analysis of gases released during charging of the Ru/B₄C cell was conducted using in-situ gas chromatography (GC). Fig. 4a shows the specific flow rate variations of O₂ and CO₂ over time during the first charging process, along with the charge voltage curve under the limited capacity of 300 mAh g⁻¹. During the whole charging process, the content of CO₂ has always been very small, indicating the dominant discharge product is Li₂O₂ in the first discharge and the side products are negligible. O₂ is generated very rapidly at the beginning of charge and its content increases almost linearly up to the maximum value of about 0.012 μmol min⁻¹ at 3.1 V. Subsequently, the O₂ specific flow rate gradually drops down to 0.008 μmol min⁻¹ at 3.4 V. From 3.4 V to 3.5 V, the O₂ specific flow rate changes smoothly, coinciding with the small change in cell voltage and indicating the charging process reaches a quasi-steady state. After 3.5 V, the O₂ flow rate decreases rapidly, corresponding to the fast increase in charge voltage. One possible reason is that most of the Li₂O₂ has been oxidized at the voltages from 3.0 to 3.5 V and mass transfer occurs due to the lack of Li₂O₂ reactant. At the same time, some insulating side products may be newly formed during charging, which cover the active sites and increase the cell impedance (Fig. S6).

In an ideal OER process where only the oxidation of Li₂O₂ occurs, the value of e⁻/O₂ should be 2.0 (see Eqn. (11)).



Thus, the theoretical O₂ flow rate was calculated based on the charge current applied, which should be about 0.0155 μmol min⁻¹. In Fig. 4a, the total O₂ specific flow rate is far less than the theoretical value, indicating part of the current is consumed by side reactions. The deviation between the actual O₂ specific flow rate and the theoretical value can then be regarded as an indication of the extent of side reactions. Some reaction intermediates (O₂^{•-} or ¹O₂) may be generated during Li₂O₂ decomposition and oxidize the electrolyte, as shown in reactions (12) and (13), rather than combine to form O₂ molecules (Eqns (14) and (15)) [10,23]. The side products caused by electrolyte decomposition may not be fully decomposed during charging and thereby accumulate in the next discharge process, leading to a vicious cycle.



Another Ru/B₄C cell prepared in the same way was deeply discharged to 2.0 V and then charged to 4.2 V alongside in-situ GC analysis

to investigate the O₂ and CO₂ content under the full discharge/charge mode. As shown in Fig. S8, the gas flow rate curve looks similar to that for the shallowly discharged Ru/B₄C cell (Fig. 4a) but the overall cell voltage is enhanced by about 0.5 V. The low CO₂ flow rate during the whole charging process indicates Li₂O₂ is still the dominant discharge product under deep discharge. More insulating Li₂O₂ products formed in the full discharge process may cause the increase of 0.5 V in the overall charge voltage.

Fig. 4b shows the O₂ and CO₂ specific flow rates during the 40th charging process. The charge voltage rises quickly at the beginning of charging, alongside the rapid generation of O₂. However, the increase of the cell voltage is much faster than that during the first charge. The O₂ specific flow rate is lower than that in the first charge and the maximum value (about 0.0085 μmol min⁻¹) occurs at about 3.4 V. Above 3.6 V, the cell voltage rises relatively slowly until the end of charging at about 4.2 V, where the CO₂ is generated rapidly while the O₂ specific flow rate keeps descending. This may indicate a large amount of side products has accumulated and the charge process from 3.6 V to 4.2 V is related to their decomposition. The in-situ GC analysis reveals that the parasitic reactions incurred by reaction intermediates during charging may cause failure to effectively refresh the Ru/B₄C electrode surface.

3.4. Degradation mechanism

Fig. 5a shows the electrode before testing has hierarchically porous structure with macropores of micrometer size and micropores of nanometer size. Fig. 5b shows the dominant discharge product has toroidal shape with a diameter of ~600 nm as shown in Fig. S9.

Fig. 6a shows the XRD patterns corresponding to each of the electrodes in Fig. 5. The XRD pattern of the Ru/B₄C electrode after the first discharge confirms that the dominant discharge product is Li₂O₂ and there are no obvious diffraction peaks related to LiOH or Li₂CO₃ (the inset in Fig. 6a). After the first charge, the toroidal particles disappear, as shown in Fig. 5c, indicating the electrode surface has been recovered well to the original morphology. Fig. 6a verifies that the diffraction peaks of Li₂O₂ cannot be found after the first charge. After 40 cycles, no toroidal Li₂O₂ can be observed (Fig. 5d). However, Fig. 6a shows the diffraction peaks of Li₂CO₃ and LiOH as well as Li₂O₂ for the Ru/B₄C electrode after 40 cycles, indicating the speed and extent of side product formation may be beyond the catalytic capability of the Ru/B₄C catalyst during charging the Li–O₂ cells. Increasing the loading of Ru/B₄C or reudcing the depth of discharge and charge may extend the cycle life of Li–O₂ batteries.

FT-IR spectra in Fig. 6b show a consistent result with XRD analysis. The FT-IR spectrum of the pristine Ru/B₄C electrode looks very similar to that of the electrode after the first charge and shows no Li₂CO₃. After 40 cycles the IR absorption peaks of Li₂CO₃ can be easily observed at

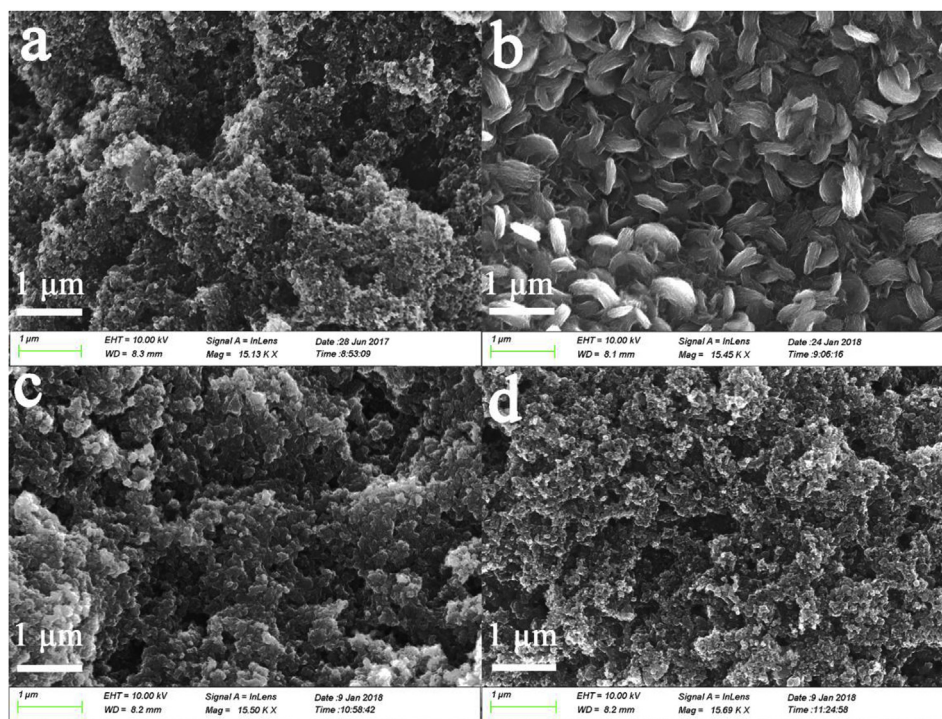


Fig. 5. SEM images of Ru/B₄C oxygen electrodes in different states: (a) before testing, (b) after the first discharge, (c) after the first charge, (d) after 40 cycles.

850 cm⁻¹ and 1443 cm⁻¹, corresponding to the vibration strengths of C–O and C=O, respectively.

Li 1s spectra in Fig. 7a show that there is a trace amount of Li₂CO₃ in the discharge product after the first discharge. The lower detection limit of XPS probably enabled it to detect the existence of Li₂CO₃, although XRD and FT-IR could not. After the first charge, the Li₂O₂ peak disappears but the peak of Li₂CO₃ still can be found, indicating the incomplete decomposition of side product on charge. The shift of the Li 1s XPS peak to the higher binding energy for the Ru/B₄C electrode after the first charge implies the formation of LiF, which may arise from the decomposition of Li⁺ salt in the electrolyte [48]. After 40 cycles, a very significant Li₂CO₃ peak and a small contribution from Li₂O₂ are shown in the Li 1s spectra, in agreement with the XRD and FT-IR results. In Fig. 7b C 1s spectra deliver a similar result to Li 1s spectra. The peaks at 290.8 and 285.0 eV represent the C–F bond of polyvinylidene difluoride (PVDF) and the C–C bond, respectively [21,23,49]. The peaks at 289.2 and 286.8 eV correspond to C=O and C–O bonds, which are from Li₂CO₃ or other side reaction products [21,23]. It can be seen from Fig. 7b that after the first discharge, there is a slight C=O peak and it

becomes a little more evident after the first charge. After 40 cycles, the C=O peak becomes significant, which matches well with the results in Li 1s spectra. O 1s spectra also exhibit the accordant result with Li 1s and C 1s spectra in Fig. 7c. B 1s spectra for the pristine Ru/B₄C electrode (Fig. 7d) shows the existence of two boron chemical states with binding energies centered at 188.4 and 190.5 eV, which correspond to B–B and B–C bonds, respectively [21]. There is no obvious change in B 1s spectra for the samples after the first discharge and charge, demonstrating the inherent stability of the B₄C substrate. After 40 cycles, B 1s features of B–B and B–C bonds are not evident because the surface of the B₄C electrode may be covered by non-boron based side products. A very slight bump around 193 eV is related to B–O surface species, indicating the surface of B₄C can be slightly oxidized during long-term cycling.

Ru/B₄C has demonstrated a powerful multifunctional activity for Li–O₂ batteries. However, the electrolyte decomposition triggered by reactive oxygen species generated during the charging process may aggravate the formation of side products. To suppress the side reactions by the use of truly stable electrolytes or additives to trap or quench the

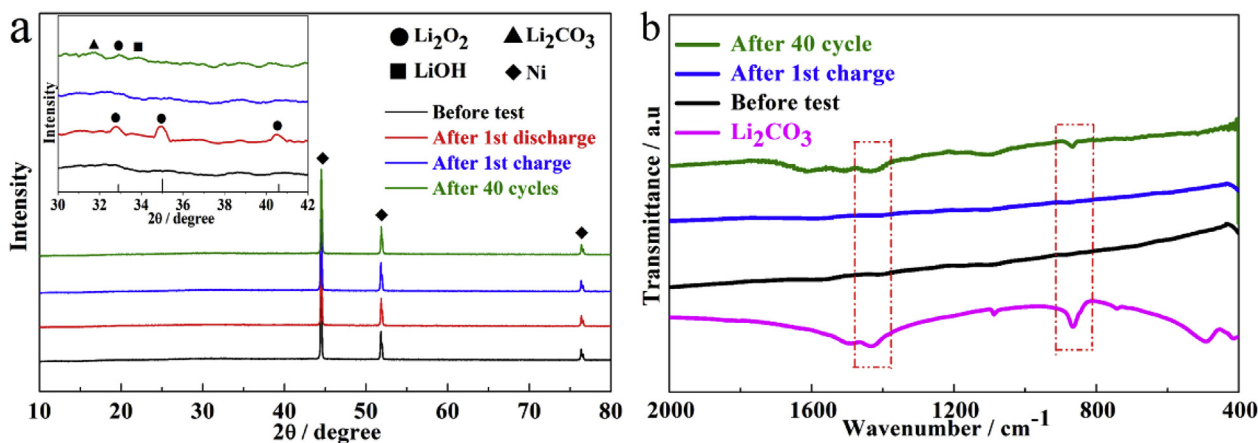


Fig. 6. XRD patterns (a) and FT-IR spectra (b) of Ru/B₄C oxygen electrodes in different states.

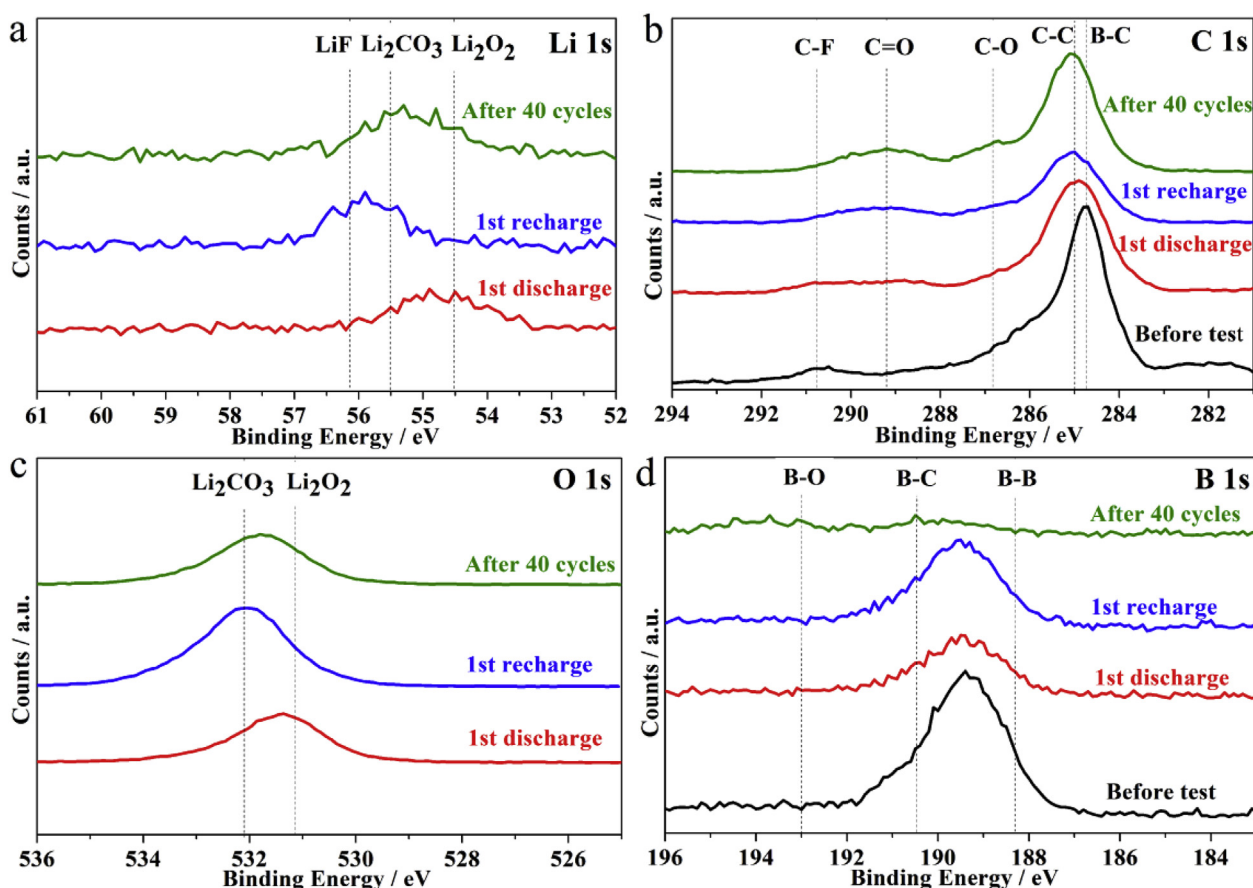


Fig. 7. Li1s (a), C1s (b), O1s (c) and B1s (d) XPS spectra of Ru/B₄C oxygen electrodes in different states.

reactive intermediates may be an effective strategy to achieve reversible Li–O₂ batteries [10]. Despite all of this, Ru/B₄C, with superior multifunctional activity for ORR, OER, and side product decomposition, can be a promising oxygen-electrode material for Li–O₂ or Li–air batteries operated in ambient air.

4. Conclusion

Ru/B₄C exhibits a superior activity for the complete decomposition of Li₂CO₃ and LiOH in the ether-based electrolyte under a significantly low voltage within the stability window of the electrolyte. A Li–O₂ battery using the Ru/B₄C oxygen electrode achieves a high discharge voltage plateau of 2.8 V, a significantly low charge voltage of 3.7 V, and excellent cycle performance under the capacity limiting protocols of 300 and 500 mAh g^{−1}. In-situ GC analysis reveals that O₂ is the major gas produced during charge. The content of CO₂ is negligible in the first charge for either a shallowly or a deeply discharged Ru/B₄C cell. Ru/B₄C demonstrates a highly efficient multifunctional activity for ORR, OER, and side product decomposition, which can be a very promising oxygen-electrode material for Li–O₂ batteries and Li–air batteries.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2018.12.016>.

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